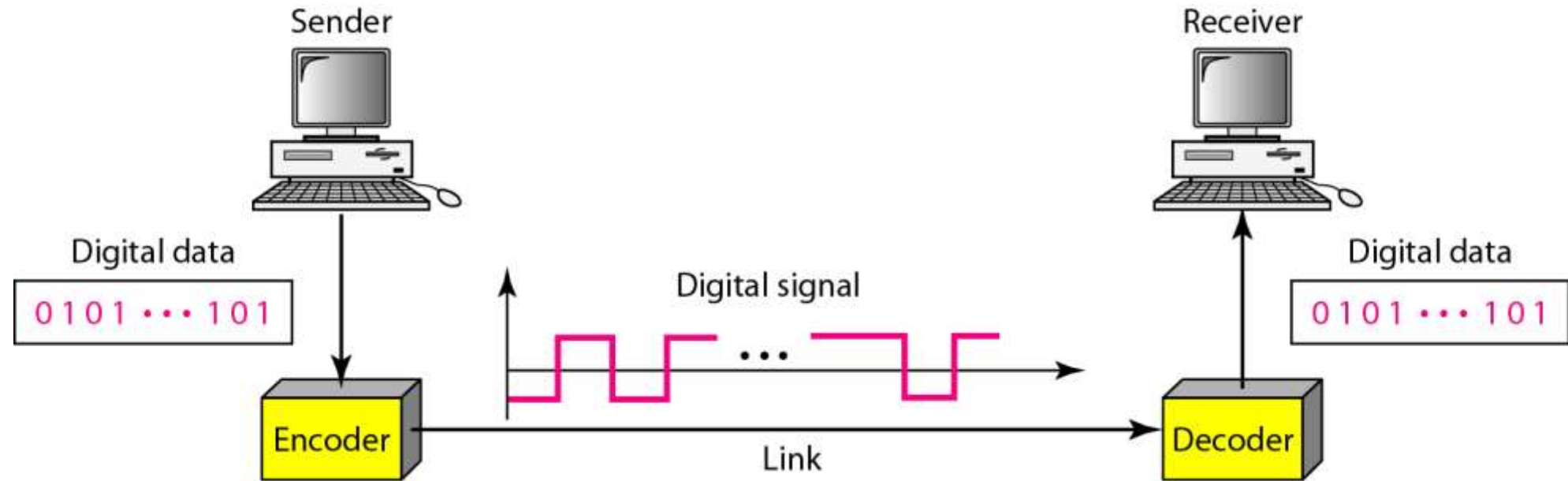


Data Communications and Networking

Chapter 2: Digital Transmission

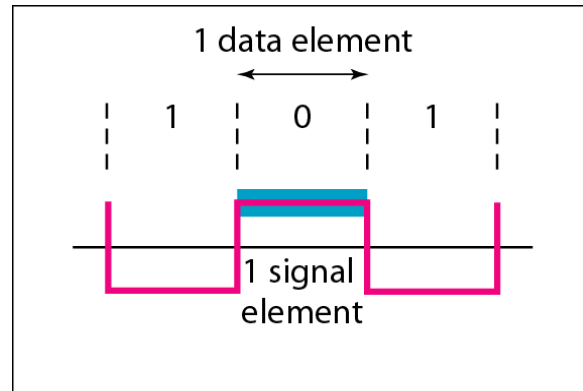
Digital Data&Signals



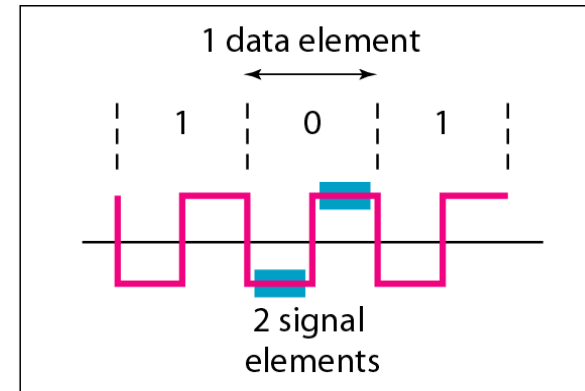
❑ **Digital data:** what we need to send, measured in **bit rate**

❑ **Digital signal:** what we can send, measured in **baud rate**

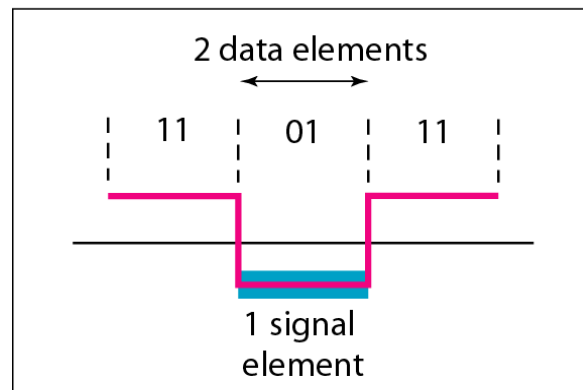
Digital Data&Signals



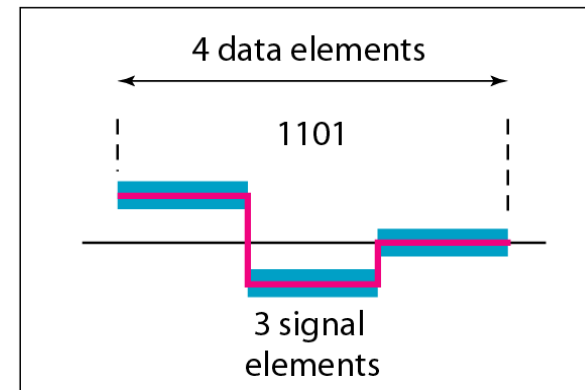
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



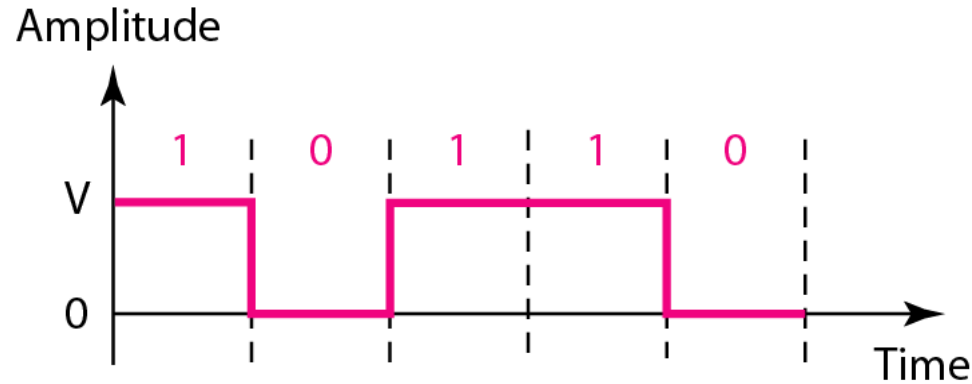
d. Four data elements per three signal elements ($r = \frac{4}{3}$)

Signal element versus data element

Digital Encoding - NRZ Scheme



Transmitting digital data over digital signals



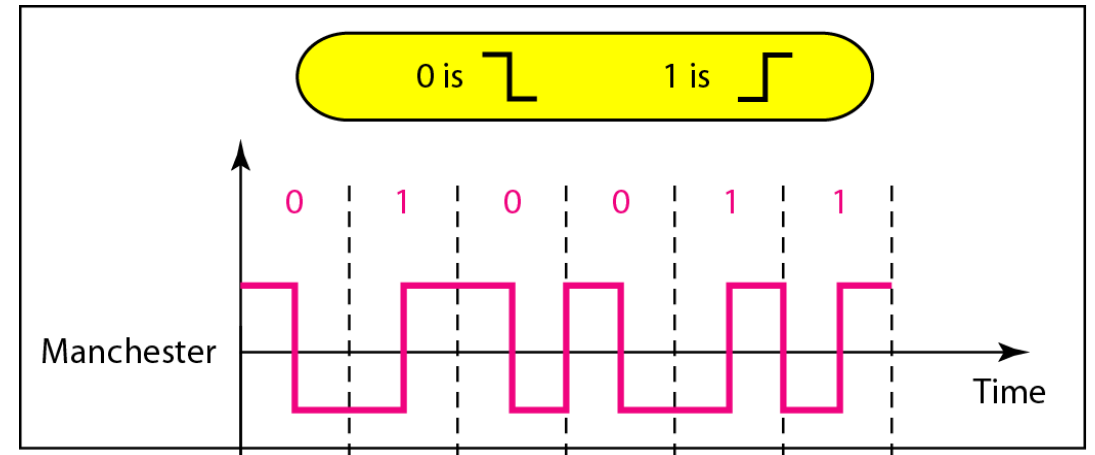
Non-Return-to-Zero (NRZ)

- the signal does not return to zero at the middle of the bit
- '1' bit is sent as a high value and '0' bit is sent as a low value.
- The receiver may lose synchronization because the data may contain long runs of consecutive bits with the same value [no changes in voltage].

Digital Encoding - Manchester Encoding Schemes

Transmitting digital data over digital signals

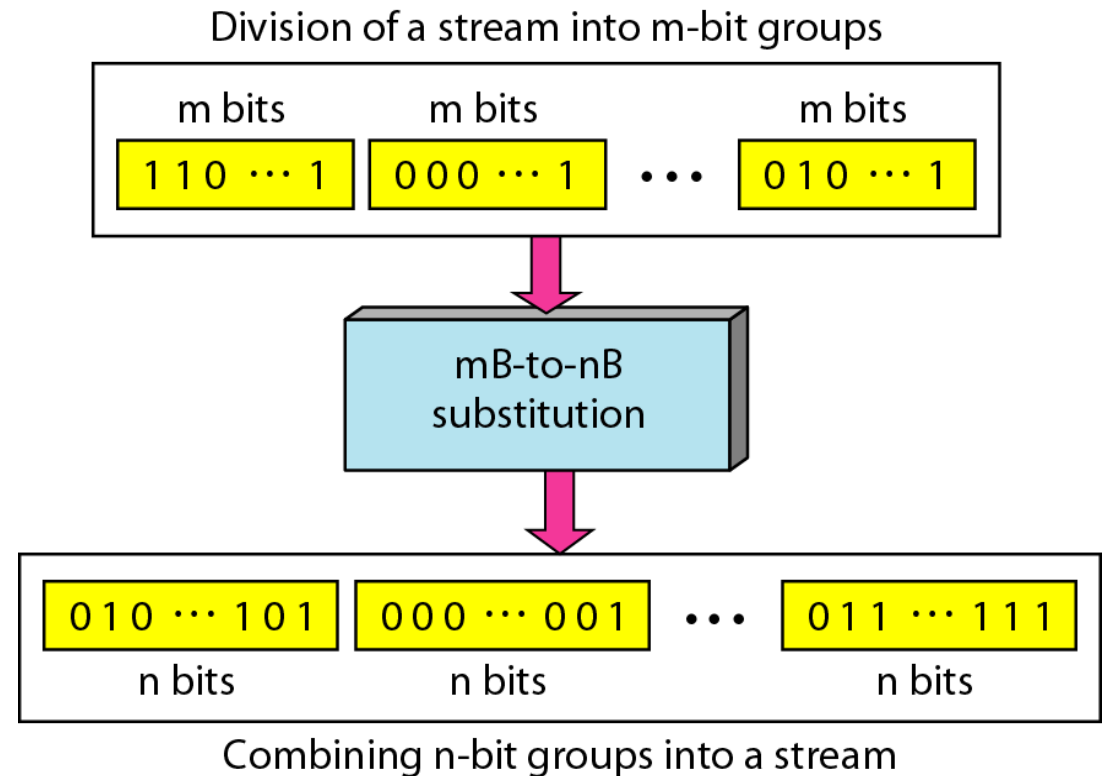
- ❑ The digital data is represented as: '1' bit by a voltage from low to high; '0' bit by a voltage from high to low.
- ❑ Manchester encoding is a synchronous clock encoding technique to encode the clock and data of a synchronous bit stream.
- ❑ The encoding efficiency of Manchester is half that of NRZ



Digital Encoding - Block Coding

Transmitting digital data over digital signals

- Block coding is normally referred to as mB/nB coding; it replaces each **m-bit** group with an **n-bit** group



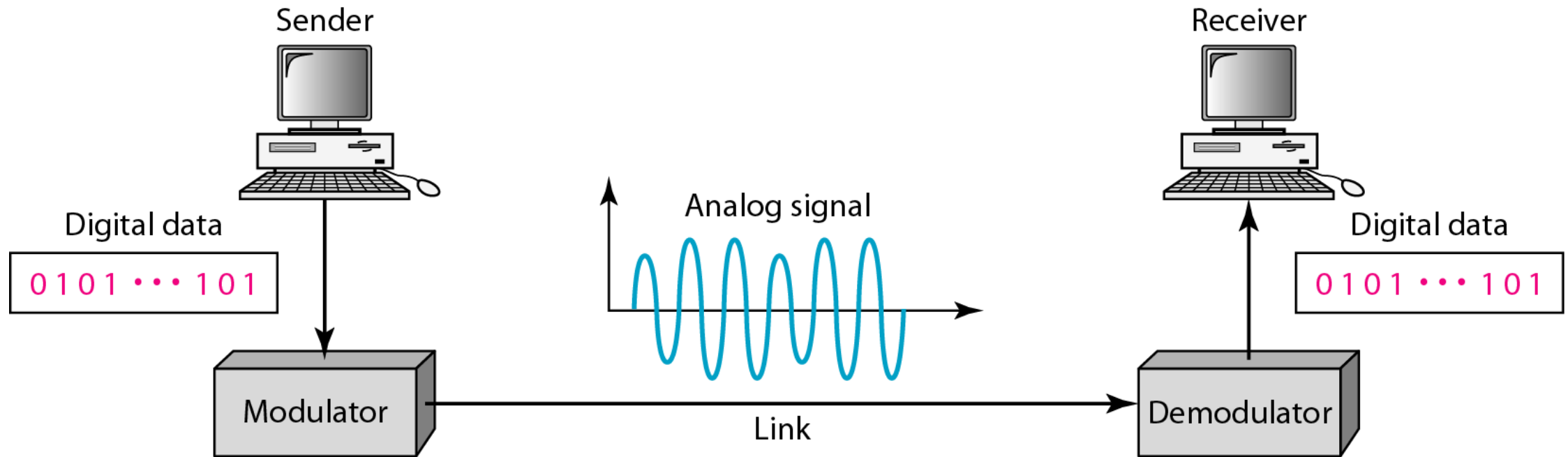
Digital Encoding - 4B/5B Encoding



Transmitting digital data over digital signals

- 5-bit output that replaces the 4-bit input
- No more than one **leading zero** (left bit) and no more than **two trailing zeros** (right bits)
- There are never more **than three consecutive 0s**
- If a 5-bit group arrives that belongs to the unused portion of the table, the receiver knows that there is an error in the transmission

Digital Data and Analog Signals



Digital Data and Analog Signals



Transmitting digital data over analog signals

□ Carrier signal

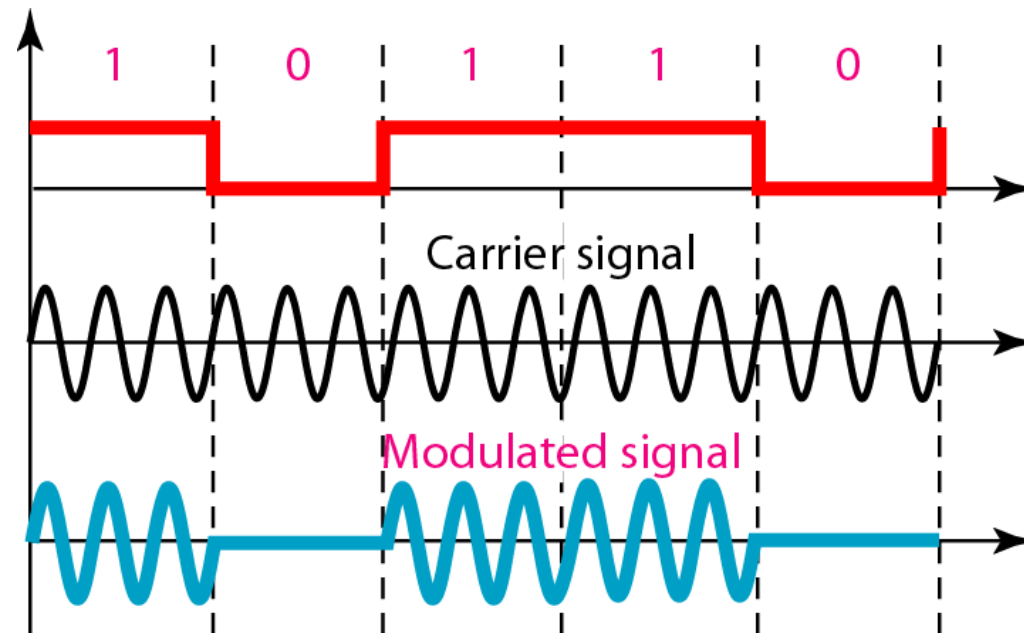
- In analog transmission, the sending device produce a high-frequency signal that acts as a base for the information signal. This base signal is called the **carrier signal** or ***carrier frequency***
- Digital information changes the carrier signal by modifying one or more of its characteristics (**amplitude**, **frequency**, or **phase**)

AM - Amplitude Modulation



Transmitting digital data over analog signals

- Amplitude Modulation modifies the amplitude of the carrier to represent 1s or 0s.
- Example: A 1 is represented by the presence of the carrier for a predefined period of 3 cycles of carrier. Absence (or no carrier) indicates a 0.

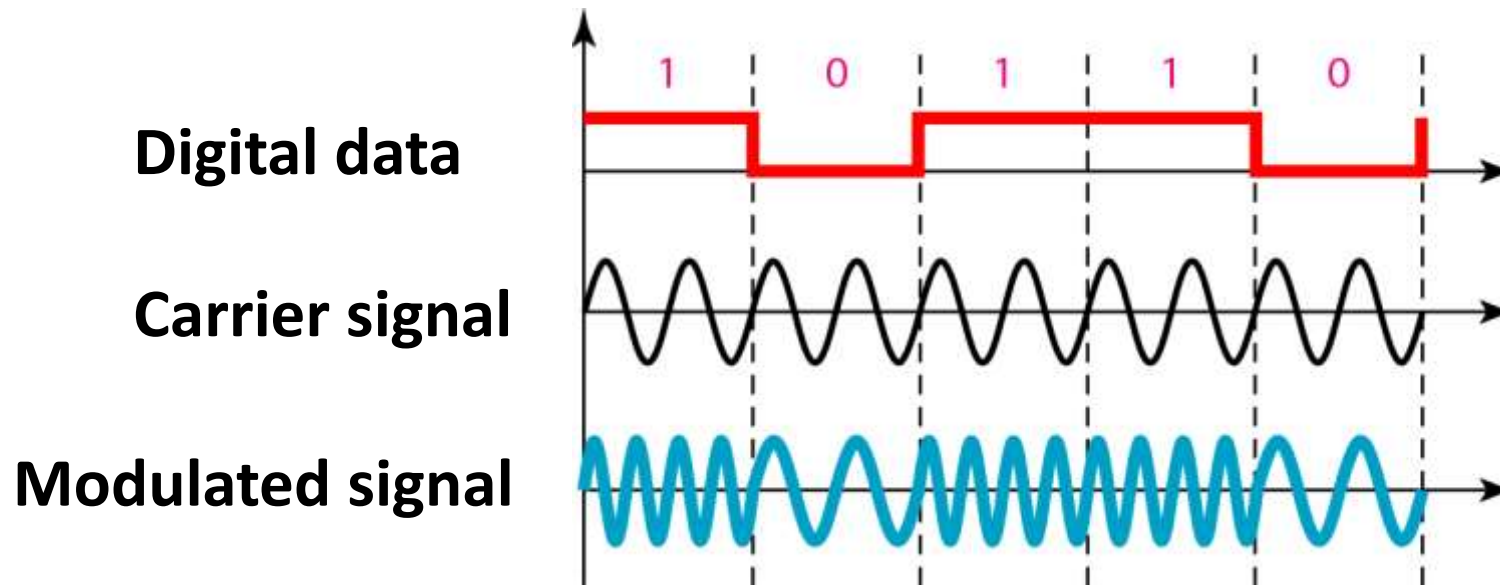


FM - Frequency Modulation



Transmitting digital data over analog signals

- Frequency Modulation modifies the frequency of the carrier to represent the 1s or 0s.
- Example: A 0 is represented by the original carrier frequency and a 1 by a much higher frequency (the cycles are spaced closer together).

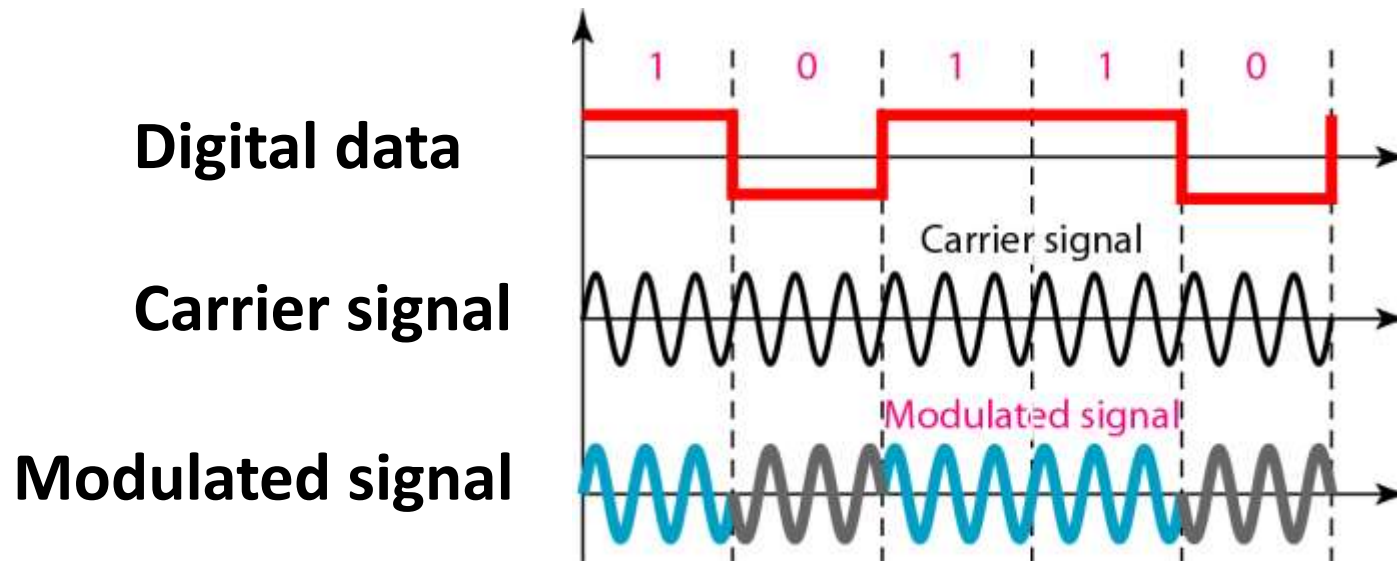


PM - Phase Modulation



Transmitting digital data over analog signals

- Phase Modulation modifies the phase of the carrier to represent a 1 or 0.
- The carrier phase is switched at every occurrence of a 0 bit, but remains unaffected for a 1 bit. The bits are timed to coincide with a specific number of carrier cycles.



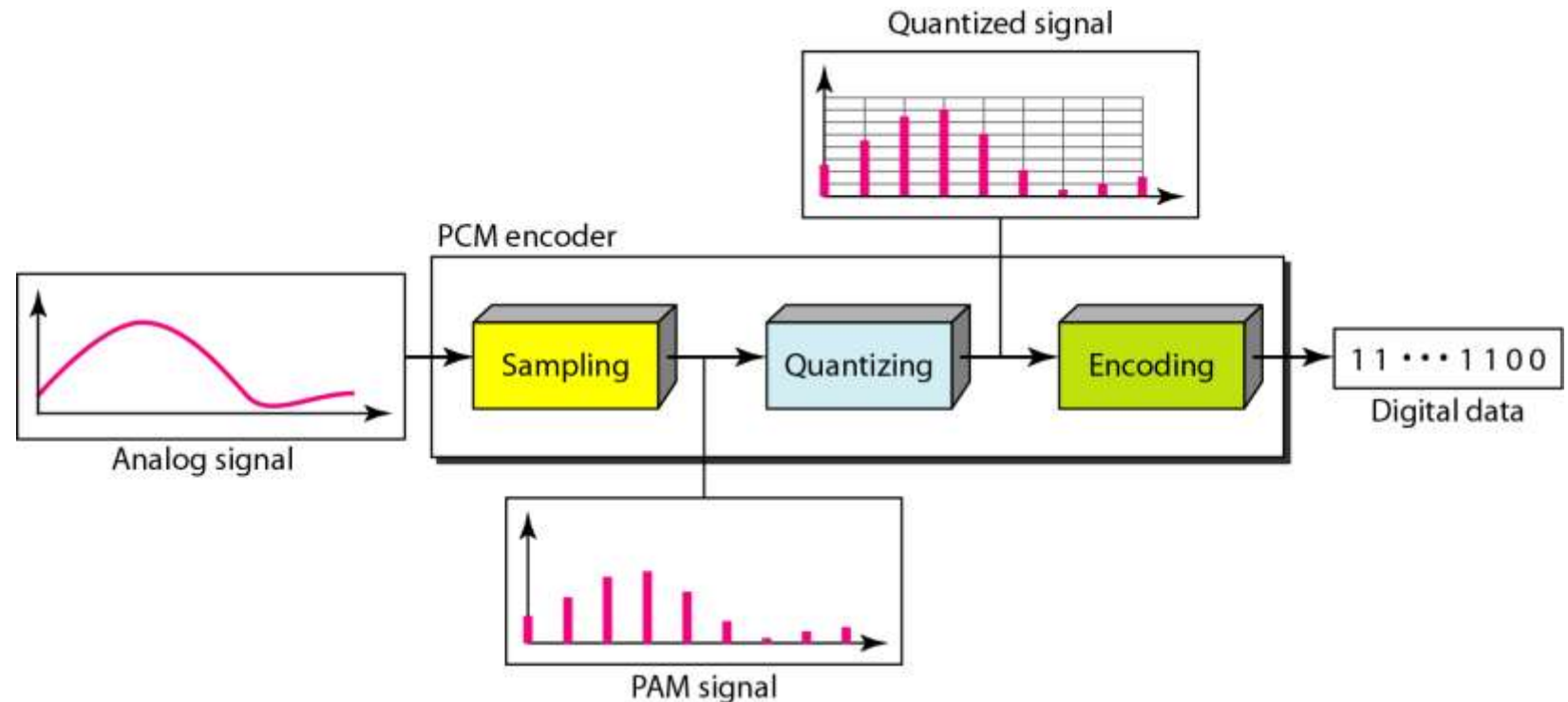
Analog-to-Digital Conversion

Convert Analog Signal to Digital Signal/Data

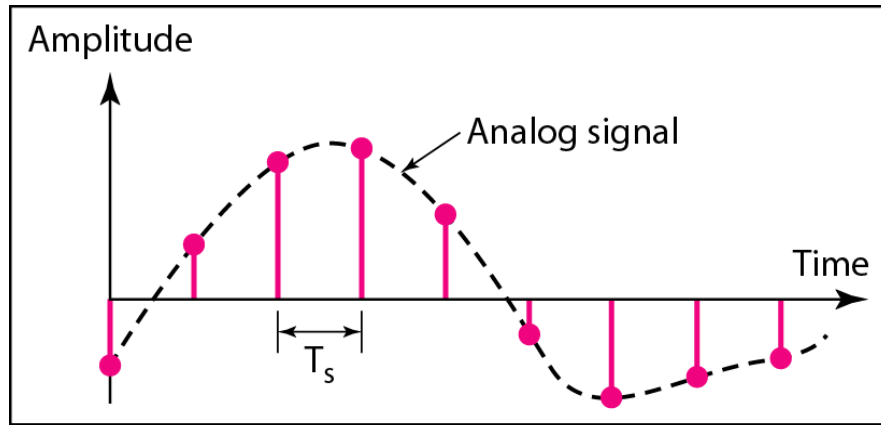


❑ Pulse Code Modulation (PCM)

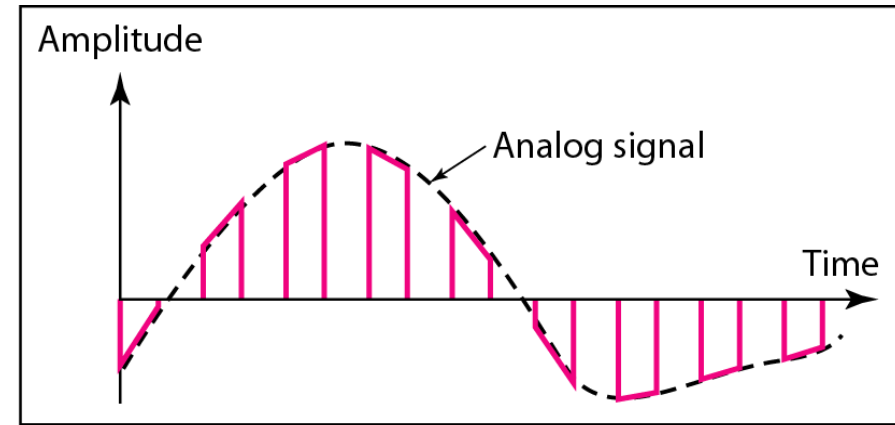
- Sampling
- Quantization
- Encoding (for digital data)



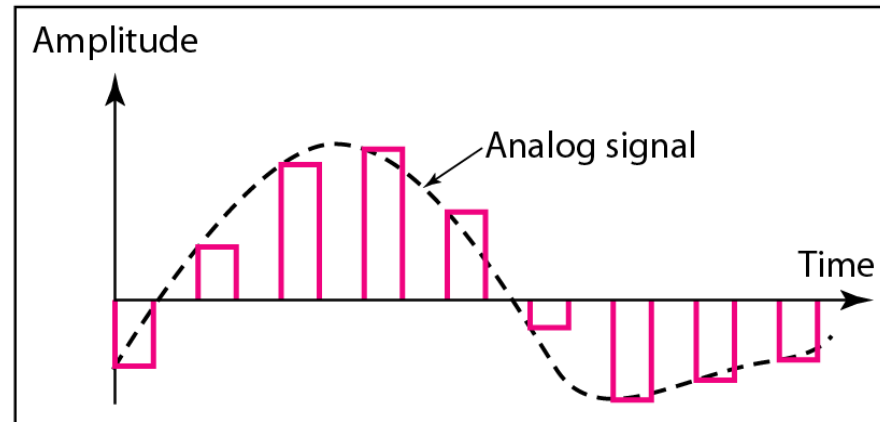
PCM



a. Ideal sampling



b. Natural sampling



c. Flat-top sampling

Three different sampling methods for PCM